

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis. Synchrony bounds content disagreement only through encoders shared on overlaps, and then only at the patch the bound is read on, not over the whole population. The resultant does not determine the state: a winding field and a uniform one are stationary at one coupling, at opposite ends of that observable. A unique fixed point is not a self-model, since a contraction can discard its input: accuracy across a family of states requires distinguishable codes. Where the encoding is a region's restriction, that resolution is demanded of the region before any readout is chosen. And agreement about a change takes communication. Four of the five are checkable for a machine candidate, which fixes what a claim of machine unity or self-representation commits its maker to measuring. Under stated hypotheses, compatible

local states then glue to a unique global state that an accurate contracting reconstruction fixes uniquely. Installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it to exceed $2D/\kappa$; κ is derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs, and an audited axiom footprint no wider than its libraries'. Further controls bound what observation can decide: distinct mechanisms share stationary phase distributions, and coherent adaptation can lose the structure it was meant to acquire. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test; the finite-size estimate here is biased below 1/2 by its escape criterion, as a deterministic reference confirms. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase

11 order. *Compatibility* concerns whether local descriptions agree where their
12 domains overlap. *Self-representation* concerns whether the resulting state
13 carries an accurate internal reconstruction of itself. We propose the latter
14 two relations as formal correlates of conscious unity and a minimal self; their
15 interpretation requires evidence beyond the phase model.

16 Unity has been approached differently elsewhere: integrated information
17 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
18 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
19 et al., 2014), and predictive-processing accounts derive both from a hierar-
20 chical generative model (Friston, 2010). Here the three requirements are
21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 9.4 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand’s relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-

34 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
 35 overlaps convert phase order into a bound on content disagreement. Compat-
 36 ible spatial measures have a unique global extension; approximate agreement
 37 controls a specified global selection. A contracting reconstruction makes that
 38 extension its unique fixed point only if it accurately reconstructs the exten-
 39 sion. Accuracy across a family of states imposes a further condition on the
 40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
 42 Stationary phase distributions can be shared by stronger coupling, weaker
 43 diffusion and a matched common drive. Finite ramps change the observable
 44 onset. A winding field is stationary at the same coupling as a uniform one and
 45 has the opposite resultant, so that observable does not determine the state.
 46 Coherent adaptation can lose alignment with a designated environmental
 47 template, and a compatibility estimator needs diagnostics that distinguish
 48 content from decoder artifacts. These findings determine the measurements
 49 needed to test the proposed connections.

50 Three of the ingredients are standard: the synchronization threshold
 51 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing
 52 of compatible local sections, and the Banach fixed-point theorem (Banach,
 53 1922). Four things here are not. The installed-energy condition bounds
 54 mean coupling by the stored energy of a declared actuator, which makes a
 55 resource condition necessary for coherence on the stationary branch. The
 56 coherence-to-content bound converts phase order into a statement about de-

57 descriptions rather than about timing, and its patch-local form is what keeps
58 it from being vacuous in a large population. A stability criterion on the
59 reweighted coupling kernel, together with an existence condition on an am-
60 plitude layer, settles when a low resultant reports weak coupling and when
61 it reports a winding field. And the composition is machine-checked together
62 with a witness discharging all eight hypotheses at once, which is what makes
63 their conjunction non-empty rather than a list no object satisfies.

64 Granting those premises, collected in Table 2, the account runs as follows.
65 Extracellular contraction on awakening raises endogenous field amplitude
66 and with it the effective coupling between cortical oscillators. Once that
67 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
68 within a quarter turn the population settles onto it; other initial data leave
69 windings stationary at the same coupling. Encoders shared where cortical
70 territories overlap turn that phase order into agreement about the quantities
71 the territories describe in common, so the local descriptions glue: there is
72 one state of which each is a restriction. Part of that state encodes the state
73 itself, and a readout of that part reconstructs it accurately enough, and
74 sensitively enough across the states that matter, to make it the unique state
75 the system models itself as being. On this account the glued state is what
76 makes distributed activity one situation, and its accurate self-reconstruction
77 is what gives that situation a perspective. Every step of it is conditional.

78 Section 2 collects the results that hold whatever physical field is proposed
79 and whatever substrate carries it; a reader who rejects the cortical hypothesis

80 of Section 8 still has to accommodate them. Sections 3–5 then develop the
 81 three requirements together (Fig. 1). Section 6 composes them, stating the
 82 thermodynamic constraints and all eight connecting hypotheses per branch;
 83 its supporting models of feedback, learning and resource supply have distinct
 84 state spaces and assumptions. Section 7 then examines what the observables
 85 can establish, before Section 8 develops the cortical hypothesis and an awak-
 86 ening experiment. The hypothesis is that extracellular contraction increases
 87 field-mediated coupling; its direction, rate and relation to recovery require
 88 independent calibration. Supplemental Material organizes the constructions
 89 and controls behind the claims they support. Table 1 lists the symbols and
 90 the section that defines each.

Table 1: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1)	Phase diffusion, in inverse time
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 7.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one

Table 1 (continued)

Symbol	Defined	Meaning
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	Section 5	Number of communication rounds to a deadline
ν	Section 9.3	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (14)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (15)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 8	Extracellular volume fraction
λ	§7.3, §8	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
E, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
$\mathcal{E}$	Eq. (13)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 7.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 7.3	Phase a winding advances across separation d
E	Eq. (15)	Field amplitude, in the cortical identification
U	§3.2, §4	Stored energy $U(\Xi)$ of a configuration; a patch U of the cover

Table 1 (continued)

Symbol	Defined	Meaning
s, s_*	§4–5	A global section and the one the reconstruction fixes

2 What holds without the cortical hypothesis

Most of what follows is conditional. Table 2 names eight hypotheses that the composition consumes rather than derives, and the cortical identification of Section 8 is a further commitment on top of them. Five results are not conditional in that way. They constrain any account that reads unity off phase coherence, whatever physical field is proposed to carry the coupling and whatever substrate carries the field. Each is stated below where it is used; collected here, they are the article’s fixed points.

Phase order alone does not constrain content. Assign every patch the same phase and let two local mass profiles disagree at a shared site. The profiles are chosen independently of phase, so synchrony bounds their disagreement not at all, and they have no common extension: perfect locking is consistent with local descriptions that do not glue (Section 4). What the positive result of Section 4.2 costs is exactly this gap. A bound on content disagreement needs encoders already related on overlaps, and that relation is a supplied fact about the descriptions rather than a consequence of the dynamics.

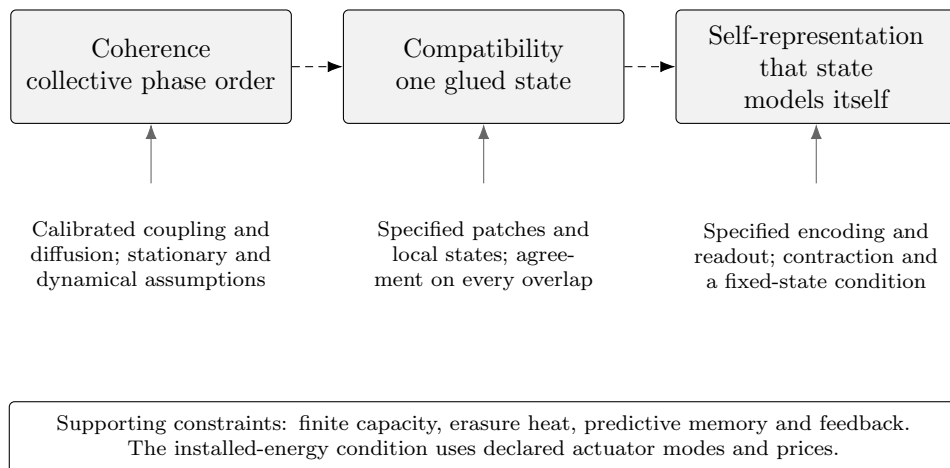


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 2 records the eight implications consumed by each branch of the formal composition.

108 **The order parameter does not determine the state.** A field wind-
 109 ing once across a periodic sheet has resultant $r = 0$ — the order parameter
 110 of Eq. (1) below — while every patch of it stays locked, and this is not a
 111 contrived initial condition. On a periodic array of any dimension, a winding
 112 is stationary for *any* isotropic symmetric kernel at identical frequencies, be-
 113 cause the coupling term is odd in the separation against a kernel even in it
 114 (`char_is_kuramoto_trajectory`). Section 7.3 exhibits a sheet on which uni-
 115 form and winding initial conditions reach 0.9657 and 0.0083 at one coupling,
 116 diffusion and kernel. A low resultant therefore reports either weak coupling
 117 or a winding field, and no measurement in r alone separates them. The state

118 also carries an integer the resultant does not: read around a ring, the same
 119 winding has winding number q , and a nonzero q obstructs every global real-
 120 valued phase lift (`winding_degree_obstructs`). The resultant is an average
 121 over sites and moves when they are reweighted or the patches redrawn; the
 122 degree telescopes and does not.

123 **A unique fixed point is not a self-model.** A constant readout con-
 124 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 125 is consequently the wrong criterion, and Section 5.2 gives the right shape
 126 of one: reconstructing M states pairwise more than 2ε apart within error ε
 127 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 128 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 129 readout whatever.

130 **Where the encoding is a restriction, the region must resolve the**
 131 **family.** Read the code as the state's own restriction to a region and
 132 the criterion moves off the readout entirely. A boundary whose region
 133 reads two relevant states identically cannot reconstruct them more than 2ε
 134 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
 135 boundary that does reconstruct a separated family restricts injectively on
 136 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
 137 region sits is fixed before an encoding is chosen, so no readout supplies
 138 resolution the region lacks.

139 **Agreement after a change takes communication.** In a synchronous
140 network whose updates read only the site’s own state and its declared neigh-
141 bours’ messages, two configurations agreeing on a site’s causal past through
142 T rounds leave that site identical at round T . Two interventions outside
143 that past that require reports more than 2ε apart therefore cannot both be
144 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
145 ity and timely compatibility are therefore different requirements, and cutting
146 the path removes the second while leaving the first intact.

147 None of these depends on the coupling being electromagnetic, on the sub-
148 strate being cortex, or on the eight hypotheses of Table 2 holding. Section 9.3
149 applies four of them — all but the one about the resultant, which is a state-
150 ment about a spatial observable — to a candidate that is neither biological
151 nor field-coupled.

152 **3 Coherence: physical coupling and its re-** 153 **source condition**

154 **3.1 Dynamics and the stationary threshold**

155 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

156 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
157 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
158 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
159 quency heterogeneity and diffusion are distinct parameters. The stationary
160 calculation below uses identical frequencies and this scalar coupling. Ap-
161 plying it to a spatial kernel requires an aggregation rule and a check of the
162 resulting approximation.

163 For identical frequencies in their rotating frame, the positive normalized
164 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

165 where I_n are modified Bessel functions and ψ is the collective phase. The for-
166 mal derivation uses the absence of stationary current for a periodic gradient
167 drift to obtain this density, then verifies normalization and self-consistency.
168 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above
169 threshold, exactly one positive solution coexists with zero and increases
170 strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

171 Figure 2 shows this stationary branch.

172 Linearization about the uniform density gives rate $K/2 - D$ for the first
173 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists

174 exactly above threshold. About the coherent density, the quadratic form
 175 is strictly dissipative transverse to collective rotation, whose rate is zero.
 176 These linear estimates concern the limiting equation. Relating them to finite
 177 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 178 tion; exchange symmetry alone does not establish the needed independence
 179 (Sznitman, 1991).

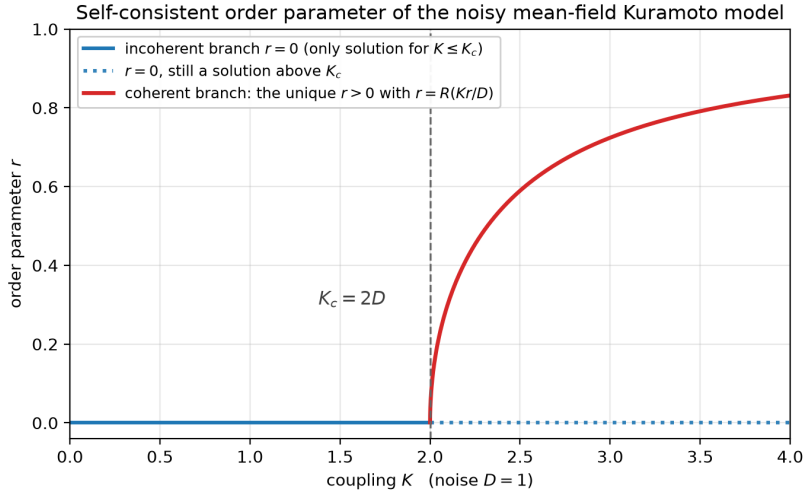


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

180 3.2 Installed energy bounds mean coupling

181 A spatial model replaces the finite sum by integration against a kernel. One
 182 declared implementation carries response modes with continuous profiles ϕ_i ,

183 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_{\Xi}(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

184 The same configuration sets the reciprocal kernel and stored energy. An
 185 executed transition changes the kernel through its changed occupancies and
 186 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 187 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 188 pair reconstruction on refining spatial partitions converges to its product-
 189 measure integral under the specified compactness and sampling assumptions;
 190 the refinement index is independent of the number of control updates.

191 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 192 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 193 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

194 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 195 coherent stationary branch and any growing Fourier mode about the uniform
 196 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 197 is necessary: energy above it need not produce sufficient coupling because
 198 the bound can be loose.

199 The quantity constrained is installed energy rather than dissipated heat,
 200 and how much the condition excludes depends entirely on κ , which the mode
 201 profiles, prices and substrate normalization supply: any macroscopic sub-
 202 strate carries installed energy far above the thermal floor, so the inequality
 203 binds through that conversion and not through the energy, and no result
 204 here derives κ for cortex or for any other candidate. What the condition
 205 contributes is a resource requirement in units a measurement can reach. Ap-
 206 plying the scalar threshold to heterogeneous or signed spatial dynamics re-
 207 quires a justified reduction, and identifying the installed kernel with a cortical
 208 field is a separate physical commitment.

209 **4 Compatibility: from coordinated activity** 210 **to a global description**

211 **4.1 Why common timing is insufficient**

212 For overlapping patches U and V , compatibility means that restricting either
 213 local description to $U \cap V$ gives the same object. Descriptions can agree on
 214 shared quantities while contributing different information elsewhere. Com-
 215 patible sections of the specified sheaf have a unique global extension. On
 216 a finite cover, local finite spatial measures agreeing as measures on every
 217 overlap also have a unique finite-measure extension whose sheaf image is the
 218 glued section. No mass-one normalization is imposed.

219 In the cup example, visual and reaching descriptions must express a
 220 shared quantity, such as cup location, in common coordinates. Restriction
 221 retains that quantity while dropping details particular to each description.
 222 Agreement permits surface appearance and hand configuration to contribute
 223 to one situation. The mathematical counterexample assigns the same phase
 224 to every patch while two local mass profiles disagree on a shared site. Those
 225 profiles have no common extension. Because they are chosen independently
 226 of phase, synchrony places no constraint on their disagreement.

227 This distinction limits the biological interpretation. Receptive territories
 228 and content domains must be related to spatial patches, and the descriptions
 229 must support the required extension. Matching overlap marginals of de-
 230 coded probability laws need not determine a unique global joint distribution
 231 (Abramsky & Brandenburger, 2011).

232 The cover is itself a choice, and the construction constrains it in one way
 233 only: synchronized patches agree where they overlap. Where they do not
 234 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 235 satisfies that obligation for *every* assignment of local descriptions and phases
 236 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
 237 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
 238 ship of the class then reports nothing about the system beyond the patches
 239 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
 240 the theorem does not assume: that distinct patches meet (`HasNonempty-`
 241 `Overlaps`), and that what they share is a quantity fixed independently of

242 the agreement it is being used to establish.

243 4.2 When coherence does constrain content

244 A declared relation between phase and content gives a positive result. Let
 245 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 246 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 247 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 248 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 249 sites; their equality on overlaps is a supplied relation between descriptions of
 250 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 251 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 252 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

253 Here discrepancy is the norm of the difference, and r and N are read on
 254 whichever population the bound is applied to. That choice is what decides
 255 whether the estimate says anything. A patch carries an order parameter of
 256 its own, and reading the bound there carries that patch's resultant and that
 257 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 258 `coherence`), so a large population observed through small patches is con-
 259 strained by the small count. Read instead on the whole array, the factor N
 260 makes the estimate weak, because it draws a pointwise guarantee from an

261 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
 262 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
 263 tion permits disagreement even at perfect locking. A common nonexpansive
 264 linear projection preserves agreement, but agreement of a projection does
 265 not establish agreement of the full vector.

266 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
 267 constrains anything only where $\sqrt{2} N \sqrt{1 - r^2}$ falls below that. The winding
 268 sweep of Section 7.3 fixes both ends of that range on one sheet. Read on the
 269 sweep's own patch, 625 sites at patch-local order 0.9015, the bound is $382.5 L$
 270 and constrains nothing; holding that order fixed, the count alone brings it
 271 below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair
 272 of the same winding, the smallest patch there is, gives $0.1388 L$. Shrinking
 273 patches therefore thins the overlaps at the same rate as it tightens the bound,
 274 and the two ends are not reached by the same cover. At a columnar grain
 275 the estimate constrains agreement inside one locally synchronous population
 276 and is vacuous across the distant territories whose agreement unity is about.

277 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 278 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 279 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 280 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

281 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-

282 duce an initial mismatch. Approximate observations leave a nondecaying
 283 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 284 update to the phase process.

285 4.3 What approximate agreement determines

286 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 287 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 288 Weights that vanish outside their patches and sum to one at each site select
 289 a global profile by weighted averaging. The selection lies within ε of each
 290 patch there, and any two such selections differ by at most ε , independently
 291 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 292 $L\varepsilon$ across selections.

293 At zero tolerance, the selection is the unique exact extension. At every
 294 positive tolerance there are families with no exact extension; selection then
 295 gives an approximation whose dependence on weights is bounded. Supple-
 296 mental Material gives a three-site witness attaining the bound. The mass-
 297 profile construction uses the scalar case of the content dynamics and addi-
 298 tionally requires nonnegativity of the initial profiles and observations. The
 299 general vector result supplies no identification with the spatial-measure sheaf.

300 These constructions concern declared objects: a specified cover, contents
 301 valued in a normed space, and encoders related on overlaps. They select
 302 neither the neural cover nor the decoder, and identifying their contents with
 303 perceptual content, or establishing that the identification bears on experi-

ence, is a further modelling task; Section 7 sets out what a measurement of it would have to establish.

5 Self-representation: a state that models itself

5.1 The reflexive construction

Compatibility gives a global state. Reflexivity relates it to an encoding within that state. Equip global sections with a complete metric. If the additional restriction-resonance condition identifies encoding with restriction to a region $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

The region and readout are fixed across the state space. If F contracts and fixes the section s_* formed from the local descriptions, Banach’s theorem makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for a relaxation time $\tau > 0$. Constructing a map with that factor remains a modelling obligation.

In the cup example, the global state includes the represented scene and the

321 perceiver's relation to it. A local encoding must retain enough information
 322 for the readout to recover that modelled state; cup position alone would be
 323 insufficient if hand configuration and viewpoint could not be reconstructed.

324 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 325 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

326 At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for
 327 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 328 weakens.

329 5.2 Reconstruction across a family of states

330 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 331 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 332 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

333 Thus M relevant states pairwise more than 2ε apart require at least M
 334 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 335 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 336 existing self-prediction map.

337 This is an additional accuracy criterion on a family, with no requirement
 338 for multiple exact fixed points. A blind encoder or constant readout fails it

339 for separated states despite having a unique fixed point.

340 Where the encoding is read as the state’s own restriction to a region, the
341 criterion becomes a condition on the region. Restriction resonance requires
342 the encoder and that restriction to identify the same pairs of states, so a
343 boundary reconstructing two relevant states more than 2ε apart while its re-
344 gion reads them identically is not resonant (`not_isRestrictionResonance_`
345 `of_reconstructs`); read forwards, a resonant boundary that reconstructs
346 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
347 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
348 cally is fixed by where the region sits, before an encoding is chosen, so the
349 resolution the family demands is the region’s and no readout supplies it. The
350 implication runs one way: a region that does resolve the family is not thereby
351 shown to carry a resonant encoding, and E89 supplies no such resolution.

352 Those two demands are jointly satisfiable with nothing folded anywhere.
353 Take the region to be the whole substrate: restriction is then the identity, so
354 the boundary is resonant, reconstructs every declarable family exactly, and
355 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
356 properness condition on the region (`IsProperAvatar`) is therefore a separate
357 requirement rather than a consequence of resonance and resolution, and what
358 a fold into a local region is asked for. Its quantitative form replaces the
359 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
360 within ε whose members lie pairwise more than $d_{\min}$ apart has δ -separated
361 codes whenever $\delta L \leq d_{\min} - 2\varepsilon$, so the family is no larger than the δ -packing

362 number of what the region carries (`encard_le_packingNumber_restrict-`
 363 `ToAvatar`). That is the shape a continuous family of states needs, and at
 364 the whole substrate it says nothing, because the region's range is everything.
 365 The gain L is declared and derived here for no physical readout, and at
 366 unbounded gain the bound says nothing.

367 **5.3 Agreement after a change takes communication**

368 Snapshot compatibility does not specify how quickly descriptions respond
 369 when a shared quantity changes. In a synchronous network whose updates
 370 read only the site's own state and declared neighbours' messages, two initial
 371 configurations agreeing on a site's causal past through T rounds produce the
 372 same state there at round T . The deadline consequence is the one Section 2
 373 states, and it follows from the reconstruction bound in Section 5.2: a report
 374 from that site is a readout of a code that the two interventions leave identical.

375 A three-site delay line witnesses failure after one round and exact report-
 376 ing after two; cutting the path prevents delivery at every round. A report
 377 may still happen to be correct about one fixed world. The bound concerns
 378 responsiveness across interventions.

379 Throughout this section *prediction* names a reconstruction map, and the
 380 relevant family of states, the metric, the tolerance, the communication graph
 381 and the deadline are all declared inputs, as are the represented variables and
 382 the fidelity counted as accurate. Reading the map as temporal neural pre-
 383 diction would require dynamics; reading the resulting code count as physical

memory capacity would require a hardware model, which a state-dependent readout would have to extend.

6 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement and reconstruction error. Thermodynamics constrains the physical operations that could implement them, and it does so from the side rather than along the spine: no edge below derives coherence, compatibility or reconstruction from a heat bound, and the resource condition these edges terminate in is necessary without being restrictive for any macroscopic candidate (Section 3.2). What this half contributes is a ledger in which a resource claim is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system–bath assumptions (Landauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result concerns a field reaching disconnected vacuum components (Kibble, 1976). The composition connects these objects through the explicit hypotheses in Table 2.

6.1 Feedback changes the information constraint

For passive memory X_t driven by an environmental signal S_t , the predictive bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

402 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
403 et al., 2012). Zero retained information also satisfies it, so a task and a
404 learning rule require additional specification. Relating this bound to Lan-
405 dauer heat requires a common thermal scale and budget, and memory waste
406 bounded by erased entropy.

407 An acting agent changes the signal it will receive. Specify policy, world,
408 observation and memory-update channels. For their finite laws,

$$\begin{aligned} I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\ &\quad - I(X_t; S_{t+1} \mid S_t). \end{aligned} \tag{12}$$

409 The second conditional term vanishes in the passive model. With feedback,
410 the signed difference can be negative as action creates correlations with the
411 future. A causal reading requires an environmental state sufficient to describe
412 the relevant dynamics.

413 For a controlled substep, the active thermodynamic bound instead uses
414 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
415 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
416 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;
417 Ito and Sagawa, 2013). Every forward path of positive probability must have
418 a positive-probability reverse for this finite identity. A missing reverse gives
419 infinite divergence. The supplement develops the support conditions and
420 composes actuation with memory updates on their shared intermediate law.

421 **6.2 Learning and repeated operation need explicit re-** 422 **sources**

423 The finite examples distinguish selecting a feasible policy, relaxing a register
424 toward an encoded reward, and learning an unknown task from observa-
425 tions. The observational learner acquires correlation with an environmental
426 parameter while its action marginal stays uniform. A continuing model exe-
427 cutes actuation, observation-driven learning and reset in sequence, preserving
428 above-chance performance at a positive work cost per cycle.

429 A phase-derived sensor makes the informational distinction concrete: an
430 informative readout and a blind control have the same phase order but dif-
431 ferent learning performance. Attaching the learner’s register to the actuator
432 of Section 3.2 ties reward, installed kernel and work to its own executed law.
433 The learning witness rises above the necessary installed-energy bound while
434 its matched control remains on the exclusion line. These finite constructions,
435 detailed in Supplemental Material, leave the evolving feedback between phase
436 dynamics, installation and sensing to be specified.

437 For that model, common energy observable $\mathcal{E}$, thermal scale $k_B T$, local
438 detailed balance and positive support at each stage give

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (13)$$

439 where expected cumulative work stays within the supplied allowance B at
440 every prefix. This bounds expectations. Separate constructions track a store

441 on each trajectory, include finite sources, charge sensor clearing and prepend
 442 preparation. They specify when a mean allowance represents an available
 443 resource; an expected budget alone permits overdrawn paths.

444 These constructions establish costs for their declared processes. A finite
 445 source with a positive net cost permits only finitely many funded cycles,
 446 while continuing operation requires replenishment. Register–bath witnesses
 447 also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply
 448 automatically.
 449
 450

451 **6.3 What the composition establishes**

452 The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 2). In the active branch, a named
 453 substep at the register’s thermal scale satisfies
 454

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

455 with B the register’s heat. The upper allocation follows from a ledger whose
 456 operations deliver that total heat to one reservoir and whose other operations
 457 do not increase joint entropy. Those ledger assumptions are physical inputs.
 458 Neither Landauer’s lower bound nor microscopic reversibility supplies them.
 459 The regime in which those premises are exhibited is a near-equilibrium one:

460 strictly positive forward and reverse elementary paths, local detailed balance
 461 at one thermal scale, and entropy differences and heats of order $k_B T$, the
 462 witnesses being two-bit systems with interaction energies of that size. Cortex
 463 is not in it. The installed energy of Section 3.2 is 2.4×10^{14} times kT at body
 464 temperature over one region and one phase-diffusion time, and a budget B
 465 of that size stands against an entropy term of order $k_B T$, so Eq. (14) is
 466 satisfied by any macroscopic candidate and excludes none of them. Reading
 467 the register instead at the scale where its premises are physical, such as a
 468 single channel, relocates the difficulty to E45Active, which would then have to
 469 carry one molecular step’s budget to the convergence of a continuum kernel.
 470 The active branch is an alternative route to the same node rather than a
 471 stronger one, and no result of Section 3.2 passes through it: the installed-
 472 energy condition is derived from E56 and E67.

Table 2: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.

Table 2 (continued)

Connection	Status	Required assumption
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours.

Table 2 (continued)

Connection	Status	Required assumption
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named

488 in one place and machine-checked together, and not that coherence follows
489 from installed energy or self-representation from coherence.

490 Joint witnesses combine a register, boundary, refining mesh and small
491 spatial cover; active witnesses add controlled thermal updates and a priced
492 installation. They establish satisfiability under their stated identifications.
493 Mesh convergence follows from refinement data, and the witnesses establish
494 no common cortical mechanism. Supplemental Material maps the claims to
495 their Lean declarations and records the counterexamples. The development
496 declares no additional axioms beyond its libraries' foundational axioms.

497 **7 What the observations establish**

498 The composition holds under its premises. Whether observation can estab-
499 lish those premises is a separate question, and the three requirements it
500 connects concern different objects: phase dynamics, local descriptions and
501 internal encodings. Their observables must retain these distinctions. The
502 following controls show which mechanisms remain indistinguishable, which
503 temporal approximations fail, which states one observable cannot separate,
504 and what independent content measurements must establish. Each is a finite
505 numerical regime on a declared model, with full parameters and controls in
506 Supplemental Material, and none of them calibrates a cortical quantity.

507 **7.1 Stationary distributions leave the mechanism un-** 508 **determined**

509 Equation (2) gives a concentration–coherence relation with no fitted shape
510 parameter. Estimate r from circular resultant length and a from density
511 shape: inverting R from the measured r would make agreement automatic.
512 Even independently estimated agreement identifies a statistical family shared
513 by different mechanisms.

514 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
515 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
516 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
517 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
518 and diffusion D_0 have that same density. Their stationary distributions co-
519 incide window by window despite different causes.

520 Resolved dynamics supply additional information. In the ideal model with
521 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
522 increment means constrain drift. Separating endogenous coupling from a
523 matched common drive additionally requires a measured drive or controlled
524 perturbation; identifying electromagnetic coupling requires evidence specific
525 to that interaction. Supplemental Material gives the equivalence construction
526 and its numerical check.

527 **7.2 Finite ramps change the observable onset**

528 The square-root law in Eq. (3) describes stationary solutions. A changing
529 coupling requires time for the phase distribution to approach those solutions.
530 In a deterministic mean-field control that crosses the threshold symmetrically,
531 none of the scanned rates meets the declared tracking tolerance: slowing
532 the crossing does not remove the terminal residual under that initialization.
533 A slow control variable therefore supplies no general guarantee of branch
534 tracking at onset. Supplemental Material gives the residual, rate range and
535 admissible rates away from threshold.

536 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
537 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
538 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
539 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
540 6 uncensored legs supply the delay scaling below; the decade-spaced ones
541 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
542 onset exponents differ from those of the exact stationary branch evaluated
543 on the same grids. At the fastest speed the ensemble mean never reaches the
544 fit window, so its fitted slope does not estimate stationary onset.

545 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
546 makes the apparent exponent depend on ramp speed and fitting window.
547 The specified deviation-based critical ramp speed is not bracketed by this
548 sweep. With fluctuating coupling, the relevant stationary comparison also
549 depends on rate: the mean coupling approximates order in the tested fast

550 regime, while averaging stationary order over coupling works for the tested
551 slow drives confined to one side of threshold. Slow crossing drives can satisfy
552 neither approximation.

553 **The prediction.** The delay scaling is what this section supports as a pre-
554 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
555 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of $[0.394, 0.494]$,
556 which does not contain $1/2$. What that shortfall measures is the escape cri-
557 terion and not the scaling. Applied to a deterministic mean-field run seeded
558 at the fluctuation floor, the same estimator returns 0.425 over the same legs
559 at the same escape criterion, inside the interval on the measurement. That
560 run has no noise to delay it, so the shortfall is the criterion's: it crosses the
561 saturation between the floor it starts from and the level it is stopped at, and
562 tightening the criterion to $r \geq 0.05$ returns 0.490. The bias is reproduced
563 where the answer is known, which is what licenses reading the measurement
564 as the law rather than as a competing exponent.

565 Stated as a law rather than as a number: onset delay scales as the square
566 root of the rate at which coupling crosses threshold. Testing it requires
567 only that $\dot{K}$ be monotonic in a manipulated rate — not its value, nor the
568 installation factor κ , nor the field-to-coupling conversion rate γ of Eq. (15)
569 — so it is a within-subject design that varies the rate of emergence from
570 anaesthesia alone, with each participant as their own control. Such a design
571 must avoid the hazard this sweep exhibits: a fixed absolute escape criterion

572 meets a size-dependent critical fluctuation floor, so the criterion has to be
 573 referred to each recording's own pre-threshold fluctuation scale rather than
 574 to a shared number. The diffusion D is the remaining input, and Section 7.1
 575 notes that it is estimable from resolved phase traces through the rate $2D$
 576 at which quadratic variation accumulates, which is a different quantity from
 577 the concentration the propofol exercise could not range over.

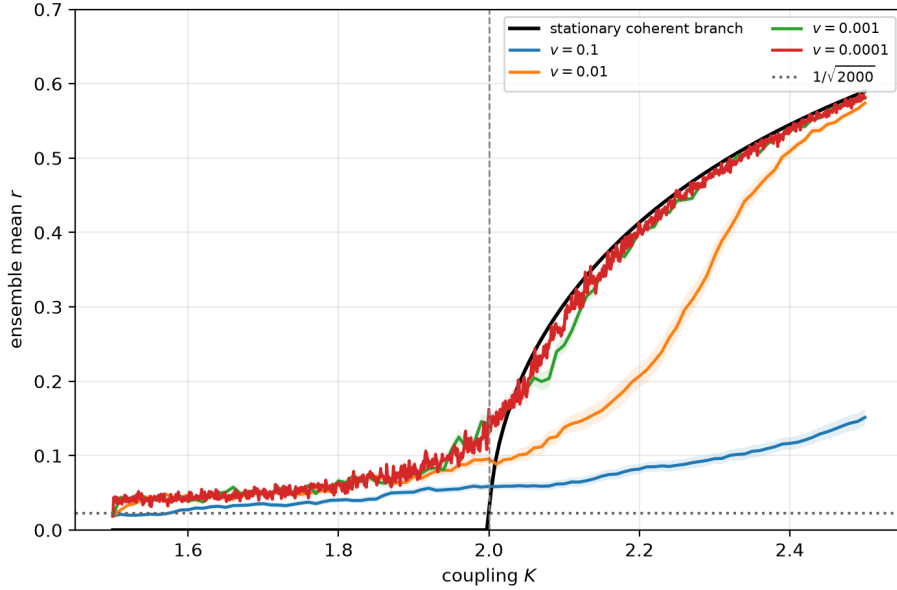


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

578 Additional controls address integration bias, population size and spatial
 579 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 580 $[1.958, 2.009]$ contains 2.

581 **7.3 A winding state leaves the order parameter am-** 582 **biguous**

583 The order parameter r is indifferent to where a phase sits in space. A field
584 winding once across a periodic sheet therefore has $r = 0$ while every patch
585 of it stays locked. Such a field is not an artificial construction here. On a
586 torus a winding is stationary for any isotropic symmetric kernel at identical
587 frequencies: the coupling term is odd in the separation against a kernel even
588 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
589 this for any finite periodic array, this sheet included. A winding satisfies the
590 positivity and identical-frequency hypotheses. It violates only the quarter-
591 turn initial spread that the convergence result additionally requires before
592 placing the limit at the uniform state.

593 On the sheet of Section 3's spatial control, a winding survives below
594 an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4).
595 That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same
596 sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a
597 uniform start and $r = 0.0083$ from a winding one. Both are stationary, with
598 patch-local order 0.9015 and no winding defect. At fixed K the observable
599 therefore takes two values spanning nearly its whole range. Coupling does
600 not determine the resultant, and a low resultant does not by itself distinguish
601 weak coupling from a winding field.

602 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-

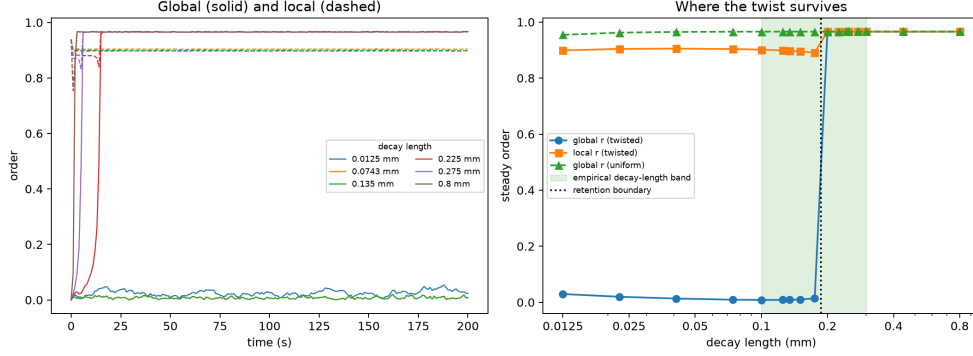


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

603 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 604 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 605 Heterogeneity shortens the range that sustains a wave without removing it.

606 Which windings persist has a criterion behind it. Linearizing the phase
 607 model about a winding returns a coupling of the same translation-invariant
 608 form, each separation reweighted by the cosine of the phase the winding
 609 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
 610 again even in the separation, so the same characters diagonalize it and ev-
 611 ery mode’s rate is one of its eigenvalues. Where the winding advances by
 612 at most a quarter turn across every separation the coupling reaches, the

613 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 614 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 615 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 616 longer separations is what carries a phase advance past a quarter turn, and
 617 that is the mechanism the decay-length sweep varies. The criterion bounds
 618 rates rather than trajectories: asymptotic stability is not proved, so the
 619 states that persist occupy a narrower range than it admits, and the bound-
 620 ary quoted above is measured and not derived from it. The same kernel is the
 621 potential's curvature, so a winding at $r = 0$ is not the global minimum the
 622 convergence result places its limit at, and one carrying a growing mode is not
 623 a local minimum either: that result's quarter-turn hypothesis must exclude
 624 a winding rather than merely decline to cover it (`char_not_potential_min`).

625 An amplitude layer decides which of these states exist, and leaves what
 626 the resultant reports unchanged. Every oscillator of the phase model sits on
 627 the unit circle, so a field with an amplitude node has no representation in
 628 those dynamics. The real Stuart–Landau extension supplies one on the same
 629 coupling, and leaves intact the gradient structure the convergence results rest
 630 on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\Gamma - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its
 631 phase equation is the Kuramoto equation with each coupling weighted by a
 632 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is
 633 this one with the amplitude ratios set to one, rather than a separate object.

634 What the added degree of freedom decides is which windings exist at all.
 635 On a periodic array of any dimension, this sheet included, a winding requires

636 $\Gamma + \lambda > 0$, with Γ the layer's linear gain. Here $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ is
 637 the eigenvalue the kernel supplies, summed over separations, with $\Delta\theta_d$ the
 638 phase the winding advances across d (`exists_charState_iff`; on the ring
 639 $\Delta\theta_d = 2\pi qd/n$). For the nonnegative kernels used here $\lambda \leq 0$, and moving
 640 coupling mass to longer separations lowers it further. A broad enough kernel
 641 therefore removes the winding outright. The sweep above reaches that same
 642 boundary by a different route: the phase model retains every winding at
 643 every coupling, and loses this one under integration instead. The two layers
 644 bound the same regime from opposite sides. A phase recording does not
 645 separate a state the amplitude layer excludes from one the dynamics fails to
 646 sustain.

647 Three limits bound the layer for a reader of the cortical literature. These
 648 states are windings and not spirals, and on the sheet that is a theorem rather
 649 than a shortage of dimensions: equal modulus at every site puts the phase
 650 circulation at exactly zero on every plaquette, and on a periodic sheet the
 651 circulations of *any* field sum to zero, so a lone defect is not a state of this
 652 geometry at all. Relaxational dynamics has no propagation speed, which is
 653 the quantity a cortical wave measurement leads with. And $|z|$ is a normal-
 654 form radius, which a measured envelope is not; this work identifies the two
 655 nowhere. Supplemental Material gives the sweep, its uniform control, the
 656 amplitude layer, the standing wave it admits inside the band, the potential's
 657 curvature at a winding and the defect theorems, together with the remaining
 658 limits of both.

659 A vanishing resultant is the absence of global agreement and not of agree-
 660 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrep-
 661 ancy bound of $2\sqrt{2}L|\sin \Delta\theta|$, with $\Delta\theta$ the phase advanced across the patch,
 662 which falls to zero as the same winding is spread over more sites (`chord_le-`
 663 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
 664 Encoders shared on overlaps therefore still bound content disagreement in-
 665 side a patch of a state that no global observable calls coherent. The winding
 666 number itself stays unconnected to content. It is, however, a property of the
 667 state rather than of the cover read on it. On a ring the whole turns dropped
 668 by the winding's affine local lifts are zero at every site but the wrap, where
 669 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
 670 every global real-valued phase lift, so one state at one coupling has resultant
 671 exactly zero and degree q (`winding_degree_obstructs`). Reweighting the
 672 sites moves the first quantity and redrawing the loop's intermediate transi-
 673 tions leaves the second where it was. The scope is the ring's: this is a loop's
 674 winding and not a spiral, the transitions are those of the affine lifts rather
 675 than of an unwrapped measurement, and the integer is identified with no
 676 cortical observable.

677 This qualifies what a phase trace can establish. An onset criterion stated
 678 in r alone registers a change of basin as readily as a change of coupling.
 679 The protocol of Section 8.3 therefore needs a spatial observable beside the
 680 resultant.

681 **7.4 Coherent adaptation can lose environmental align-** 682 **ment**

683 An adaptive coupling rule might supply representational structure, but its
684 objective must reward the relevant agreement. For symmetric coupling, pair-
685 wise sine terms cancel in the population mean drift. Minimizing squared drift
686 at that fixed mean therefore rewards equalizing drift across nodes. In the
687 frequency-cluster control, this favors coupling between unlike-frequency clus-
688 ters, whereas the diagnostic environmental template rewards coupling within
689 clusters.

690 The formalization proves descent of the squared-drift objective for a spec-
691 ified gradient flow at fixed phases. Its functional differs from thermodynamic
692 entropy production: at a coherent stationary density the probability current
693 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
694 rial gives the operator, its gradient and the conditions of the descent result.

695 The finite control co-evolves noisy phases and symmetric, nonnegative
696 couplings at fixed total resource. Its template is a diagnostic covariance of
697 latent signals, never a training input. Across three seeds, second-half order
698 remains between 0.9167 and 0.9222, while the within-cluster to between-
699 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
700 Random-update, permuted-gradient and frozen-kernel controls are reported
701 in Supplemental Material.

702 The immediate post-update objective reduction disappears when aver-

703 aged over complete update intervals. A prespecified grid of 6 rate-cadence
704 combinations contains no case meeting the 5% sustained-reduction criterion
705 while retaining coherence and losing alignment. The finding is coherent adap-
706 tation away from the specified structure, with no demonstrated sustained cost
707 reduction.

708 Kernel alignment and overlap compatibility are different observables.
709 This control addresses acquisition of its designated template under its par-
710 ticular objective. Learning models can explicitly include data in an objective
711 (Corberi et al., 2026); an account of agreement about the cup must specify
712 the content its learning rule is meant to preserve.

713 **7.5 Content agreement needs independent decoders**

714 A neural test needs independently identified domains, local descriptions and
715 restriction maps. Two decoders should estimate a shared content on over-
716 lapping domains before their agreement is examined. Discrepancy requires
717 a null under decoding noise and controls using deliberately compatible and
718 incompatible examples. An implementation on constructed data separates
719 these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by
720 trial; the compatible construction scores 0.476 on an interval containing one
721 half. Shared-prior shrinkage can nevertheless manufacture agreement, and
722 regional bias can manufacture disagreement. Per-patch decoding diagnostics
723 detect these failures; the compatibility statistic alone does not.

724 Phase agreement cannot replace that measurement. On constructed con-

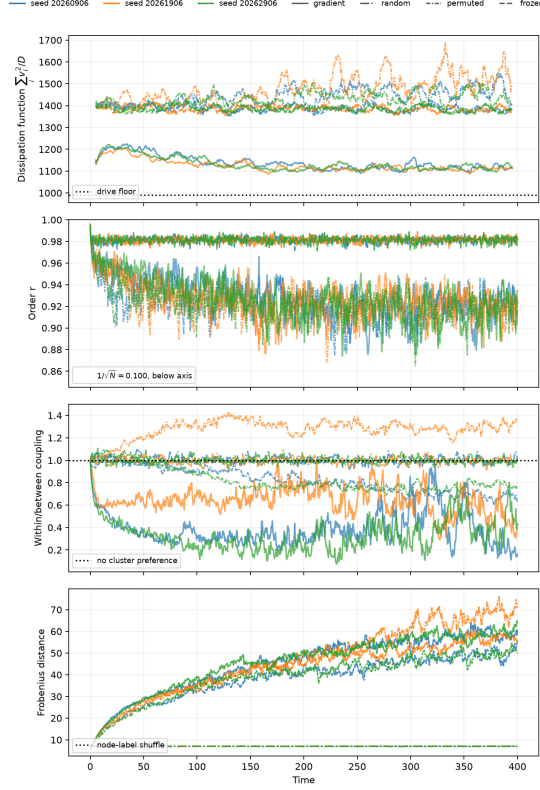


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

725 tent, phase carries a share 1.000 of the content entropy when content is built
726 from it and 0.000 when it is not. A phase-derived statistic reports com-
727 patibility on phase-locked patches whose local descriptions disagree. The
728 implemented content statistic detects their incompatibility. These controls
729 validate a measurement on known constructions; a neural content model and
730 decoder remain to be established.

731 Choosing the cover also chooses a candidate subject: trivial descriptions
732 can agree while carrying little content, and a partition can conceal disagree-
733 ment. The useful choice must therefore be constrained by what neural ac-
734 tivity encodes. The exact composition assumes compatibility of its supplied
735 cover. The content mechanisms above identify additional hypotheses under
736 which agreement can arise; they leave the empirical choice of content and
737 the connection to that cover explicit.

738 **7.6 What an informative phase-density measurement** 739 **requires**

740 The concentration range and the dependence structure decide whether the
741 concentration–coherence relation can be tested at all. Under the specified
742 log-density estimator and bias-floor criterion, discrimination at the observed
743 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
744 designs, rising to 31000 when sites share a mean in groups of 10. These
745 counts describe that estimator and dependence model; the intended recording

746 requires its own calibration. Supplemental Material reports the bin-count
747 controls and the full dependence-aware design.

748 That ceiling comes from exercising the estimators on public propofol-
749 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
750 Bipolar and common-reference montages produce different residuals. The
751 bipolar results are compatible with the Bessel relation over a concentration
752 range too narrow to distinguish it from its linear approximation at the ob-
753 served scatter, which is the range the design scan takes as given. Temporal
754 pooling changes the estimand, and sedation recordings leave the awakening
755 hypothesis untested.

756 An informative dataset therefore needs sufficient concentration range or
757 precision, calibrated estimators and an observation model for dependence,
758 source mixing and reference choice. Public mouse calcium imaging provides
759 thousands of spatially located neurons across waking, sleep and anesthe-
760 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
761 estimation band used here. Site count, state coverage and bandwidth are
762 separate requirements.

763 **8 The cortical hypothesis and an awakening** 764 **test**

765 The cortical proposal identifies a physical source of coupling and asks whether
766 its changes help explain recovery. The tests in Section 7 determine what

767 observations must distinguish before a phase trace can support this identifi-
768 cation.

769 8.1 Physical identification and calibration

770 The cortical hypothesis identifies part of effective coupling with endogenous
771 extracellular electromagnetic interactions. Fields can modulate spike timing
772 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
773 propose coordination of distributed neural information (Pinotsis & Miller,
774 2022; McFadden, 2020). Phase organization under propofol motivates study-
775 ing this regime (Bastos et al., 2021), but does not measure coupling.

776 The formal field predicate requires a jointly continuous kernel, a normal-
777 ized substrate, coupling equal to the kernel average, and no single site pro-
778 viding more than half that strength. Toy constructions satisfy it. A cortical
779 realization must specify the scale at which the kernel and its reduction ap-
780 ply: calcium imaging reports different state-dependent modularity patterns
781 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

782 Cortical columns are the obvious candidate unit, and they answer one
783 of the two demands the formalism makes. As the oscillator index they are
784 a serious proposal: they are delineated by response properties and anatomy
785 rather than by the agreement they would be used to establish (Mountcastle,
786 1997), and a phase can be estimated from a column-sized territory at the
787 electrode spacings used for cortical wave measurements (Townsend et al.,
788 2015). As the content cover they are a poor one, for three separate reasons.

789 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
790 the compatibility obligation without constraining anything (Section 4). The
791 cover condition additionally requires the patches to exhaust the substrate,
792 while much of association cortex has no accepted columnar parcellation. And
793 the column's standing as a canonical cortical unit is contested (Horton &
794 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
795 supplies the oscillator index and leaves the content cover to be identified on
796 separate evidence, which is the bridge assumption E78 records.

797 The spatial reduction is also testable. A row-sum aggregation at zero
798 diagonal tracks the uniform-coupling control over the tested spatial decay
799 range, while frequency heterogeneity displaces the operational threshold. The
800 fit window biases both estimates relative to $2D$; agreement with the control
801 is therefore the supported conclusion. Supplemental Material reports the
802 reduction, its numerical error and the distinct assumptions needed for the
803 kernel-average identification in Eq. (5).

804 The field commitment requires physical evidence beyond a choice of math-
805 ematical representation, and the available support comparisons do not supply
806 it. A finite matrix spread over positive-mass cells preserves its total cou-
807 pling and phase correlation exactly, while point-supported vanishing under
808 an atomless measure and a disadvantage from excluding a better-correlated
809 pair concern specified support constraints; Section 9.3 reads the same com-
810 parisons against a digital candidate. Neither supplies energy prices.

811 A spike-time shift alone supplies no coupling rate. The dimensionless

product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (15)$$

This rate describes how induced timing shifts accumulate into drift. Neither it nor the installation factor κ follows from a heat bound. Testing the physical threshold requires independent estimates of the interaction kernel, its aggregation and diffusion in the same cortical regime.

8.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (15) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

831 For a smooth crossing at nonzero speed, evaluating the stationary branch
 832 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

833 This follows from Eq. (3) at fixed D and concerns the stationary branch
 834 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 835 dently justified tracking approximation. The crossing controls in Section 7.2
 836 leave that approximation unsupported at onset, even though a separation of
 837 geometry and phase time scales can support it away from threshold. The
 838 awakening hypothesis therefore needs a calibrated dynamical prediction for
 839 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 840 does not supply a recovery latency.

841 **8.3 A protocol for the awakening experiment**

842 The physical test has six requirements:

- 843 1. **Coupling and diffusion in the same units.** Estimate them from re-
 844 solved phase dynamics, separating heterogeneity and observation noise.
 845 Measure the drive or perturb the system to distinguish endogenous cou-
 846 pling.
- 847 2. **Geometry calibrated to coupling.** Validate the geometry observ-
 848 able and its relation to effective coupling, with uncertainty. The con-

ductivity argument supplies only the expected direction.

3. **Calibrated phase measurements.** Estimate r by resultant length and a by unweighted regression of log density on centered cosine. Choose site count and binning from a dependence-aware calibration over the intended concentration range. The tested designs require thousands of sites at the observed concentrations; temporal pooling changes the spatial estimand.

4. **An independent recovery endpoint.** Measure behavioural recovery separately from phase organization.

5. **A predeclared dynamical onset criterion.** Set a coherence threshold above the finite-size floor and evaluate its latency under the measured coupling trajectory, including ramp speed or fluctuations. Record a spatial observable beside the resultant, since a winding field reports a low resultant at coupling that sustains local order (Section 7.3).

6. **Matched model comparison.** Compare coupling increase, diffusion change, common drive and alternative recovery models under the same observation and filtering assumptions.

Failure of the concentration–coherence relation over an informative range constrains the density model. Failure of the calibrated geometry–coupling association constrains the physical gate. Better prediction of independent recovery by a competing mechanism counts against its proposed role. A

870 favorable outcome still leaves the content of coordinated activity to be ex-
871 plained.

872 **9 Discussion**

873 Two things have been established here, and they are of different kinds. The
874 constraints of Section 2 hold outright: synchrony does not deliver compatible
875 content, the resultant does not determine the state, a unique fixed point is
876 not a reconstruction, a region that cannot resolve a family cannot encode it,
877 and agreement about a change takes communication. Any account that reads
878 unity off phase coherence has to meet them, this one included. The account
879 set out in Section 1 is conditional at every step by contrast: each connection
880 in it is a hypothesis supplied rather than a result derived, and Table 2 names
881 all eight. Its four non-standard ingredients, named in Section 1, are the
882 installed-energy condition, the patch-local form of the content bound, the
883 criterion separating a winding field from weak coupling, and the machine-
884 checked composition with its joint witness. What follows takes the conditions
885 in turn, as commitments the proposal incurs, as measurements that would
886 bear on them, and as questions the framework leaves open.

887 **9.1 Unity and a minimal self as empirical commit-** 888 **ments**

889 We propose the state glued from compatible local descriptions as a corre-
890 late of conscious unity, and its accurate internal reconstruction as a correlate
891 of a minimal self. The motivations are that different contents belong to
892 one experienced situation and that the situation is experienced from a per-
893 spective. The conditional results identify relations a realization would need:
894 sufficient installed resources for the specified phase model, agreement about
895 shared content, and an encoding that supports reconstruction at the required
896 fidelity. Discharging E12–E89 for cortex would make the composition a state-
897 ment about cortex: that a cortical field above threshold relaxes to a cover
898 whose compatible local descriptions glue, and that the glued state is the
899 unique fixed point of a reconstruction map realized there. It would leave the
900 two identifications above exactly where they are, because neither is among
901 the eight and neither follows from their conjunction; they are the commit-
902 ment this subsection names, and Section 9.2 lists the three observations that
903 would count against it.

904 Compatibility allows different descriptions to contribute different infor-
905 mation. It also permits uncertainty and conflicting judgments: apparent cup
906 position and judged cup position can be distinct quantities in a consistent
907 description of their discrepancy. Which quantities are shared must be de-
908 termined independently. Choosing them solely to secure agreement would

909 deprive the proposal of empirical force.

910 The minimal-self proposal concerns the perceiver’s position and condition
911 within the situation, such as the cup’s location relative to a body that can
912 reach it. The encoding could be implicit and need carry no autobiograph-
913 ical narrative. Its proposed necessity is philosophical: even an unreflective
914 episode has a perspective. Gluing by itself requires no self-encoding, and re-
915 constructability alone supplies no functional role. Altering the encoded hand
916 or viewpoint while holding the external scene fixed should change estimates
917 of the perceiver’s relation to the cup as the readout predicts. Whether that
918 represented perspective is experienced remains a further question.

919 **9.2 What a test of the full proposal must establish**

920 Both uniqueness results concern specified objects. They choose neither the
921 cover nor the encoding, and do not select between candidate subjects. In-
922 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
923 here selection requires independent evidence about neural organization. A
924 constant readout can contract and fix a predetermined state while discard-
925 ing its input. The family criterion in Section 5.2 tests the sensitivity that
926 this construction lacks, provided the relevant states and fidelity are declared
927 independently.

928 A test must separately measure coupling relative to diffusion, content
929 compatibility and self-reconstruction, alongside an independent endpoint for
930 experience. Meeting all three without a unified episode would challenge suffi-

931 ciency. A unified episode with incompatible descriptions would challenge the
932 unity identification; one lacking the specified self-encoding would challenge
933 the reflexive requirement. Awakening, dreaming and dissociative states offer
934 possible settings, but absence of report cannot by itself define absence of
935 experience.

936 The next physical requirement is a calibrated geometry–coupling relation
937 and resolved phase dynamics. The installation bound contributes a nec-
938 essary resource condition where its hardware representation applies; cortical
939 mode prices and field-mediated coupling require measurement. For the repre-
940 sentational connection, an informative experiment asks whether a controlled
941 change in coordination alters independently decoded content agreement, with
942 controls for sensory information and decoder quality. The constructed-data
943 estimator establishes feasibility for specified contents. Neural local descrip-
944 tions, shared overlap encoders and the self-encoding have yet to be identified.
945 The population-size dependence of the phase-to-content bound further limits
946 its quantitative use, in the form Section 4.2 reads off the sheet.

947 The separation results explain why these measurements must be inde-
948 pendent. A stationary distribution leaves several recovery mechanisms indis-
949 tinguishable. Equal phase order can accompany incompatible descriptions
950 or different task information. The tested plasticity rule preserves coherence
951 while losing its designated template. Each finding concerns the specified
952 model and observable; their role is to delimit the evidence needed for the
953 proposed connections.

9.3 What the conditions constrain in a digital candidate

The results of Section 2 hold for any candidate, and for a large language model (LLM) running on graphics processing units (GPUs) four of them are checkable rather than merely stateable. Those four are where this framework bears on claims made for such systems. It does not exclude them, and no result here could; what it does is fix what a claim of unity or self-representation commits its maker to measuring.

Codes, not reports. Reconstruction across a family is a counting constraint before it is anything else. If two macrostates the candidate is said to represent satisfy $E(s) = E(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out reconstructing both within ε , whatever the readout. A system’s reports are readouts. Identical verbal reports across two conditions are therefore consistent with identical internal codes and settle nothing about the states, while the converse claim — that the system represents its own condition across a declared family — asserts that its internal code separates that family, which is a statement about activations under intervention and not about text. Naming the channel makes the count concrete. Where the code is the token sequence a model emits—which is what its next step reads—the alphabet is the deployed vocabulary (`tokenChannel`), and a family reconstructed through it is no larger than the vocabulary size raised to the number of steps (`card_le_card_tokens`). Over an extended exchange that number is large and the

bound is slack; at a single step it is one vocabulary’s worth, so a claim that one emission reports the model’s own condition across a declared family asserts that the vocabulary separates that family (`card_le_card_tokens_one`). A claim resting on activations rather than on text has declared a different channel, to which the same count applies with that channel’s alphabet. The measurement follows from the statement: declare the family and the fidelity first, then test whether the encoding region distinguishes its members.

The region, not the readout. Where the encoding is read as the state’s restriction to a region of the machine, the demand falls on the region before any readout exists. For a decoder-only transformer the causal mask makes the network an instance of the update model of Section 5: the activation at position t and layer ℓ reads positions at most t at layer $\ell - 1$ and nothing else, so a region’s causal past is fixed by the architecture rather than chosen by interpretation. That instance is formal, and what it establishes is sharper than slow growth: the causal past of position t stays inside the positions at most t at every depth (`mask_ball_subset_le`), so the deadline bound of Section 5 fires at every deadline at once rather than at a chosen one, with its communication graph given rather than assumed (`mask_no_guarantee`). That is a statement about one forward pass; across steps a model does read its own prior output, which is the channel the preceding paragraph counts. Read against the coupling proposed here the asymmetry runs both ways, and neither way is an advantage. A kernel of full support has no neighbourhood

998 structure for the bound to bite on—after one round the causal past is the
 999 whole substrate, so the hypothesis that fires the obstruction is unsatisfiable at
 1000 any deadline but zero (`not_outside_past_of_isFullSupport`)—but a physi-
 1001 cal field propagates at a finite speed, so treating one that way is a calibration
 1002 against the phase time scale and not a theorem. What the comparison does
 1003 fix is where the bound has force. Not on the attention graph, where a position
 1004 reads every position at or before it and the fan-in is the context length. On
 1005 the physical interconnect, where fan-in is genuinely bounded: a graph whose
 1006 sites each hear from at most ν others reaches at most $(T + 1)\nu^T$ sites in
 1007 T rounds (`card_ball_le_mul_pow`), so a model sharded across many devices
 1008 pays a number of rounds growing like $\log_\nu$ of the device count for guaranteed
 1009 agreement, and buying that deadline back costs wiring—reaching every site
 1010 at a single round requires a device hearing from all but one of the others
 1011 (`le_degree_of_reaches_one`). Both ν and the device count are declared, as
 1012 the communication graph and the deadline already are, and what the count
 1013 bounds is the graph a guarantee requires rather than what hardware is build-
 1014 able. A claim that a model reports on its own processing consequently carries
 1015 an obligation to say which positions and layers the report’s causal past con-
 1016 tains, and a report that appears to track an intervention outside it indicates
 1017 a confound or a different region, not a faster channel.

1018 **Agreement about what.** Unity requires the candidate’s local descriptions
 1019 to agree on independently identified shared quantities (Section 4). Common

1020 timing, a shared residual stream and fluent output supply none of that, for
1021 the reason Section 2 gives: coordination bounds content disagreement only
1022 through encoders already related on overlaps, and which quantities two parts
1023 of a model describe in common is a decoding question carrying the failure
1024 modes of Section 7. The cover is a choice, and choosing it to secure agreement
1025 empties the claim.

1026 **What is not excluded.** Digital representations can satisfy compatibility,
1027 so finite hardware is compatible with the gluing construction and none of
1028 the above is a no-go for machines. The coupling-gated route E56–E78 is a
1029 different matter: it requires a calibrated physical coupling, its noise regime
1030 and a connection from coordination to content agreement, and an attention
1031 matrix or a shared clock supplies none of these identifications. Computing a
1032 field model likewise leaves its realization in the computing substrate to be es-
1033 tablished. Those physical obligations fall on the cortical proposal identically.
1034 The capacity comparison those support comparisons lack is available, and it
1035 favours neither side by construction. A continuum code space read at resolu-
1036 tion δ holds at least its volume divided by the volume of a resolution cell, so
1037 it out-resolves an alphabet of 2^b codes exactly when that volume exceeds 2^b
1038 cells (`two_pow_lt_packingNumber_of_lt_measure`); below that threshold the
1039 inequality runs the other way, and at a coarse enough resolution the region
1040 holds a single code. What decides the comparison is the volume, the res-
1041 olution and the dimension, all measured quantities, and continuity decides

1042 nothing on its own: a continuum code space read at a finite resolution is
1043 a finite alphabet whose capacity is a bit count. For the same reason this
1044 account credits an analog medium with no computation beyond a Turing
1045 machine's. Real-valued states carry unbounded capacity only at unbounded
1046 precision, and the noise floor $D > 0$ that Eq. (1) is stated against removes
1047 it.

1048 The installed-energy condition of Section 3.2 is restricted in the same
1049 way. It constrains a macroscopic candidate only through κ , which is hard-
1050 ware data supplied by mode profiles and prices, and Supplemental Material
1051 reads the condition against a cortical metabolic budget to return the conver-
1052 sion factor it would require rather than a verdict. The hardware comparisons
1053 there have similarly restricted scope: fixed wiring can impose a correlation
1054 disadvantage, and point-supported kernels vanish under an atomless contin-
1055 uum measure, but neither comparison makes the representation conditions
1056 unattainable on GPUs. A universal exclusion would need a necessary condi-
1057 tion for consciousness and a proof that every relevant implementation violates
1058 it, and the present composition supplies neither.

1059 Whether experience requires a living substrate is the explicit subject of
1060 biological naturalism (Seth, 2026a,b). On that view the properties ground-
1061 ing experience are those of self-sustaining organisms, and simulating them
1062 does not secure them. The conditions stated here name no biological mech-
1063 anism, but computation described as such does not satisfy them either: a
1064 candidate must exhibit a physically realized coupling in a measured noise

1065 regime, agreement about independently identified shared quantities, and an
1066 internal reconstruction meeting a declared fidelity. Whether a realization
1067 of those conditions that is not alive would be accompanied by experience
1068 is the point at issue, and these results do not settle it. A related position
1069 locates the difference in the mode of computation rather than in life as such:
1070 biological systems compute across scales that do not separate, and along-
1071 side discrete operations carry continuous-valued dynamics belonging to their
1072 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
1073 here is one measurable instance of that requirement, fixed to a physical field
1074 in a declared noise regime, and a description of such a field does not satisfy
1075 it.

1076 The neuroscientific case against *human-like* consciousness in text-only
1077 LLMs concerns embodied sensory interaction, thalamocortical organization
1078 and biological regulation linking perception and action to survival (Aru et al.,
1079 2023). Their absence supports skepticism, conditional on their causal rele-
1080 vance to experience. Those mechanisms are not requirements of the abstract
1081 gluing and reconstruction definitions, and workspace-like representations re-
1082 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
1083 investigate. What this framework adds to that investigation is an order for
1084 the questions: declare the family of states and the fidelity, fix the encoding
1085 region and read its causal past off the architecture, and only then ask what
1086 the system reconstructs. A claim that survives those three steps is a claim
1087 about the machine; one that does not survive them was a claim about its

1088 output.

1089 **9.4 Explanatory scope**

1090 Integrated-information, global-workspace and predictive-processing accounts
1091 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
1092 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
1093 candidate physical interaction and conditional relations between local content
1094 and self-encoding. Gluing and reconstruction permit multiple substrates;
1095 the electromagnetic commitment enters through physical calibration. That
1096 permission is wider than the unconditional results of Section 2 alone. The
1097 conditional chain is substrate-neutral in the same way: its field condition
1098 asks for a jointly continuous kernel on a normalized substrate, a coupling
1099 equal to that kernel’s double average, a noise that is the field’s own, and
1100 no single site carrying more than half the coupling—and none of these dis-
1101 tinguishes an electromagnetic kernel from any other continuous mean-field
1102 kernel. The site condition is the only one with physical flavour, and it holds
1103 vacuously wherever the substrate has no atoms (`no_site_dominates`), which
1104 is the continuum case the electromagnetic hypothesis is about. The chain
1105 therefore transfers to extracellular ion diffusion, to gap junctions modelled
1106 as a density, or to a mechanical or optical medium, modulo the calibration of
1107 Section 8.1. What carries the electromagnetic identification is that calibra-
1108 tion together with the extracellular-geometry argument of Section 8, which is
1109 the one place a field equation enters. The formalization checks implications

1110 under their hypotheses, and its witnesses establish realizability on declared
1111 models. Two things stand outside that result and are not supplied by it: the
1112 eight hypotheses are discharged for toy components rather than for cortex,
1113 and the identification of the glued state with unity and of its reconstruction
1114 with a minimal self is a further commitment that the eight do not contain.

1115 The explanatory question is whether the specified relations occur in neural
1116 activity and matter for experience. In the cup example, this means establish-
1117 ing how coordination helps maintain one represented situation and how the
1118 perceiver’s place within it contributes to the episode. The proposal identi-
1119 fies candidate correlates and tests of that identification. Why those relations
1120 would be accompanied by experience remains unexplained.

1121 **Data and code availability**

1122 The Lean development, simulation scripts, tests and saved numerical sum-
1123 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
1124 `cortical_coherence`. Publication simulation values are generated from
1125 saved summaries without rerunning production sweeps. The exploratory
1126 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
1127 processing and selection are described in Supplemental Material. The repos-
1128 itory contains the reproducible analysis and does not redistribute the raw
1129 recordings.

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1131 This research received no external funding. The author declares no compet-
1132 ing interests.

1133 **Use of AI tools**

1134 AI assistance to this work was substantial and is not separable by task. Sev-
1135 eral coding assistants and chat interfaces were used in an interleaved way
1136 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1137 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1138 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1139 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1140 accessed with the Hermes CLI. Between them they contributed to the Lean
1141 development, the simulation and analysis code, and the drafting and revision
1142 of both documents. Because the tools were used in combination and in alter-
1143 nation, no section, proof, figure or result is attributable to any one of them,
1144 and none is claimed to be. The author is responsible for all content, including
1145 everything a tool produced. Machine checking establishes the stated Lean
1146 results under their hypotheses; it does not validate biological interpretations
1147 or replace scientific review of generated code and prose.

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